

Dear Editor-in-Chief,

I am pleased to submit my manuscript

“When descriptive forecast-horizon gains do not imply global evidence of incremental meteorological predictability: a multi-site PM₁₀ study”

for consideration for publication in *Stochastic Environmental Research and Risk Assessment*.

The manuscript is original, has not been published previously, and is not currently under consideration by another journal.

A recurring difficulty in environmental forecasting is that a result that looks convincing descriptively may support a much narrower conclusion once the source of the improvement is examined statistically. This is particularly relevant in multi-horizon forecasting. A model may outperform persistence over a longer sequence of lead times after new predictors are added, yet the longer skill horizon does not by itself establish that those predictors contribute incremental predictive information. Our manuscript examines this distinction for hourly PM₁₀ forecasting and asks a deliberately focused question: when meteorological observations extend persistence-relative forecast skill, how much evidence supports attributing that extension to the additional meteorological information?

We study Madrid Casa de Campo and eight monitoring stations in Ireland using an expanding-window rolling-origin evaluation over January–July 2023, with forecast origins spaced by 24 hours and horizons from 1 to 24 hours. Persistence and SARIMA provide temporal reference models, while the controlled information-set comparison uses the same direct XGBoost learner first with lagged PM₁₀ and then with the addition of meteorological observations available at the forecast origin. This paired design allows the descriptive effect of the added information to be examined without changing the learner. It also keeps the interpretation deliberately retrospective: the meteorological inputs are observations available at forecast issuance, not future numerical weather prediction forecasts.

The empirical results initially suggest a strong meteorological benefit in Madrid. Under the primary definition of the useful forecast horizon—the longest contiguous run of positive skill relative to persistence—the horizon increases from 9 to 17 hours when meteorological observations are added, an 8-hour extension. The corresponding relaxed last-positive-skill horizon, however, increases from 15 to 17 hours, a 2-hour change. The contrast becomes still less straightforward across Ireland: the mean strict horizon changes from 21.875 to 22.875 hours (+1.000 hour), and five of the eight stations already reach the 24-hour evaluation boundary with the lags-only model, leaving little observable room for the strict horizon to extend further. These results make the descriptive story interesting, but they also make clear why the number of additional “useful” hours cannot by itself answer the attribution question.

That question produces the main inferential turning point of the paper. Across the 36 planned station–horizon comparisons, all tests are valid under the dimensionally corrected overlap-based DM-HLN specification, but none survives global Bonferroni correction. Limited within-station signals remain under station-wise false-discovery-rate control. We then address the most immediate statistical objection: although the 24-hour spacing between forecast origins implies no mechanically required positive-lag covariance from overlapping forecast errors for the tested horizons, residual serial dependence in the loss differential may still be present. A fixed automatic Newey–West sensitivity analysis allows for such positive-lag dependence. The family-wise result remains the same—again, no comparison survives global Bonferroni correction—although the identities of the limited local signals are not identical across the two specifications. The complete cell-level results are supplied as Online Resource 1, alongside the main-text DM-HLN inferential landscape (Figure 3).

For us, this distinction between the descriptive and inferential results is the main value of the study. The Madrid skill extension remains a genuine empirical feature of the forecasts; the inferential analysis does not make it disappear. Rather, it changes what can reasonably be claimed from it. Persistence-relative horizon gains, local evidence of forecast differences, and family-wise support for incremental predictive information are different levels of evidence. Treating them as interchangeable can make an environmental forecasting result appear more general than the evaluation actually supports.

We believe this question is well suited to *Stochastic Environmental Research and Risk Assessment* because it concerns not only predictive performance but the stochastic structure and statistical interpretation of environmental forecasts. The comparison is conducted under temporal dependence, across multiple sites and horizons, and against persistence—a demanding reference when environmental concentrations are strongly autocorrelated. The study also illustrates a practical methodological consequence of the evaluation design: when forecast horizons are expressed in physical hours but forecast origins are more widely spaced, the dependence parameters used for forecast-comparison inference must be defined on the loss-differential sequence rather than transferred mechanically from physical time.

The manuscript therefore does not argue that meteorology is generally uninformative, nor that a particular forecasting architecture is universally superior. Its contribution is more specific: in this multi-site PM₁₀ experiment, substantial descriptive horizon gains can coexist with limited family-wise evidence for attributing those gains to the added meteorological information, and that global conclusion persists under a fixed sensitivity to additional serial dependence. We believe that this provides a useful framework for interpreting horizon-dependent performance claims in environmental forecasting more generally, while keeping the empirical conclusions bounded by the evaluated sites, period, information sets, and 24-hour forecast range.

Code, processed research materials, and figure-generation resources associated with the study are made available through the project repository described in the manuscript.

Thank you for considering this manuscript for publication in *Stochastic Environmental Research and Risk Assessment*. I would be grateful for the opportunity to have the work considered by the journal’s editors and reviewers.

Yours sincerely,

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